

Day 3 – Salesforce Interview Preparation

Part 1: Interview Warm-Up

1. What is a Validation Rule?

A Validation Rule is a Salesforce feature that verifies whether the data entered into a record follows predefined business rules before the record is saved. If the rule evaluates to TRUE, Salesforce blocks the save operation and displays an appropriate error message.

Example:

Prevent a student from submitting a job application if their CGPA is lower than the minimum CGPA required for the job.

2. What is a Flow?

A Flow is Salesforce's declarative automation tool that allows you to automate business processes without writing Apex code. Flows can perform actions such as creating, updating, deleting records, sending emails, and executing logic based on specific conditions.

Example:

Automatically populate the Application Date whenever a new application is submitted.

3. What is an Apex Trigger?

An Apex Trigger is a block of Apex code that runs automatically before or after database events such as insert, update, delete, or undelete. Triggers are used when the required business logic is too complex for declarative tools like Flows.

Example:

Prevent duplicate job applications or update related records after an interview is completed.

4. When would you use a Flow instead of an Apex Trigger?

A Flow should be used when the automation is straightforward and can be implemented without code. Since Flows are declarative, they are easier to create, maintain, and modify.

Common Use Cases:

- Automatically populate field values
 - Send email notifications
 - Create or update related records
 - Automate approval or notification processes
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5. Can a Validation Rule update another field? Why?

No.

A Validation Rule is designed only to validate data before a record is saved. It cannot modify or update field values. Its only purpose is to either allow the record to be saved or prevent the save by displaying an error message.

To update fields automatically, use either a Flow or an Apex Trigger.

6. What is the execution order of Flow, Validation Rule, and Trigger?

For record creation or update, the simplified execution order is:

1. Before-Save Record-Triggered Flow
 2. Before Trigger
 3. Validation Rule
 4. Record is saved
 5. After Trigger
 6. After-Save Record-Triggered Flow
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7. What is a Record-Triggered Flow?

A Record-Triggered Flow automatically runs whenever a record is created, updated, or deleted. It is commonly used to automate business processes without writing Apex code.

Example:

When a student submits a new application:

- Automatically populate the Application Date.
 - Send an email notification to the Placement Officer.
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Part 2: Business Scenario

Scenario

You are developing a Placement Management System. Select the most suitable Salesforce automation tool for each requirement and explain your choice.

Requirement 1: Notify the Placement Officer when a student submits an application.

Solution: Flow

Reason:

An After-Save Record-Triggered Flow can automatically send an email whenever a new Application record is created. This is a standard automation task that does not require Apex.

Requirement 2: Automatically populate the Application Date.

Solution: Flow

Reason:

A Before-Save Record-Triggered Flow can assign today's date before the record is saved. This approach is faster and more efficient than using Apex.

Requirement 3: Prevent duplicate applications for the same student and job.

Solution: Apex Trigger

Reason:

An Apex Trigger can query existing Application records before insertion. If a duplicate application is found, it can stop the transaction using `addError()`.

Requirement 4: Reject applications when the student's CGPA is below the minimum requirement.

Solution: Validation Rule

Reason:

A Validation Rule checks the CGPA before saving the record. If the student's CGPA is below the required value, Salesforce prevents the record from being saved and displays an error.

Requirement 5: Automatically create an Offer Letter when the application status becomes "Selected."

Solution: Flow

Reason:

A Record-Triggered Flow can monitor changes to the Status field and automatically create an Offer Letter record when the status changes to Selected.

Summary Table

Requirement	Solution	Reason
Send email notification	Flow	Automates email notifications without code
Auto-fill Application Date	Flow	Updates the field automatically before saving
Prevent duplicate applications	Apex Trigger	Handles complex duplicate checking and validation
Reject low-CGPA applications	Validation Rule	Validates business rules before saving
Create Offer Letter	Flow	Automatically creates related records when the status changes

Part 3: Design Challenge

Requirement	Recommended Solution	Explanation
Automatically populate the Application Date when a new application is created.	Flow	A Before-Save Record-Triggered Flow fills the Application Date before the record is saved.
Prevent students with insufficient CGPA from applying.	Validation Rule	Validates the CGPA before saving and displays an error if the requirement is not satisfied.
Prevent duplicate applications for the same student and job.	Apex Trigger	Queries existing records and prevents duplicates using <code>addError()</code> .

Requirement	Recommended Solution	Explanation
Send an email to the Placement Officer after a student submits an application.	Flow	An After-Save Record-Triggered Flow automatically sends the notification.
Automatically create an Offer Letter when the application status changes to "Selected."	Flow	Detects the status update and creates the related Offer Letter record automatically.
Update multiple related records with complex business logic.	Apex Trigger	Best suited for advanced processing, bulk operations, and complex business requirements.

Part 4: Hands-on Task

Objective

Create a Record-Triggered Flow for the Placement Management System that performs the following actions:

- Automatically populate the Application Date.
- Send an email notification to the Placement Officer.
- Verify that the automation executes successfully.

Implementation

Step 1: Automatically Populate the Application Date

Flow Type:

Before-Save Record-Triggered Flow

Object:

Application

Trigger Condition:

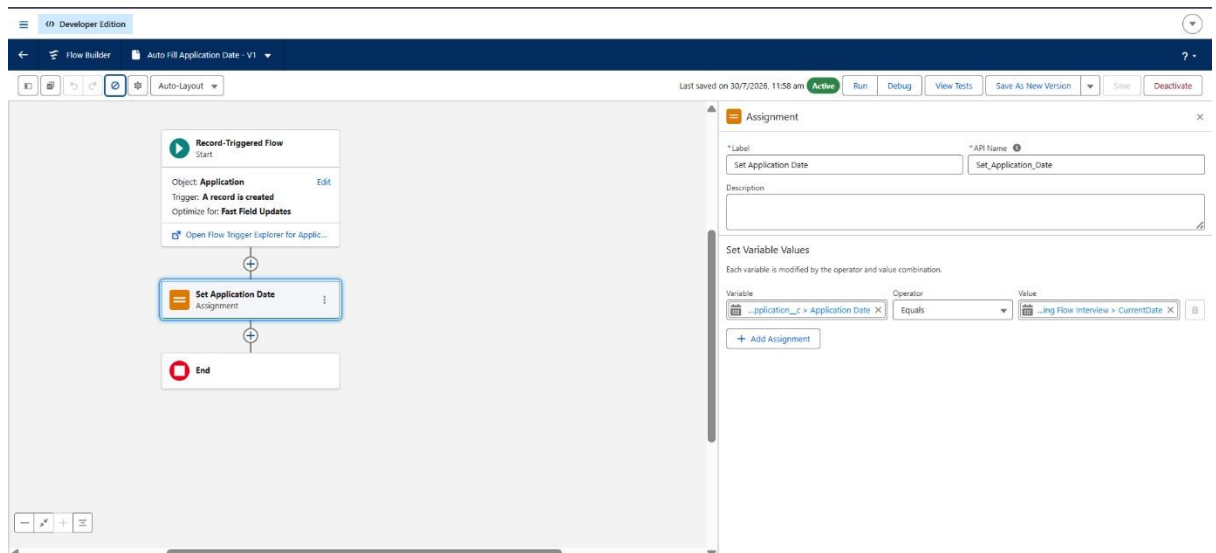
When a record is created

Assignment:

Application_Date__c = \$Flow.CurrentDate

Outcome:

Whenever a new Application record is created, the Application Date is automatically set to the current date before the record is saved.



Whenever a student submits a new application, the Application Date is automatically set to the current date.

The screenshot shows the Salesforce Playground Starter interface. The application name is REGD00000000003. The student is Jaya, the job is Salesforce Developer, and the application date is 30/07/2026. The status is Applied. The created by and last modified by are both Jaya Sri Padilam, with the last modified date being 30/07/2026, 12:01 pm.

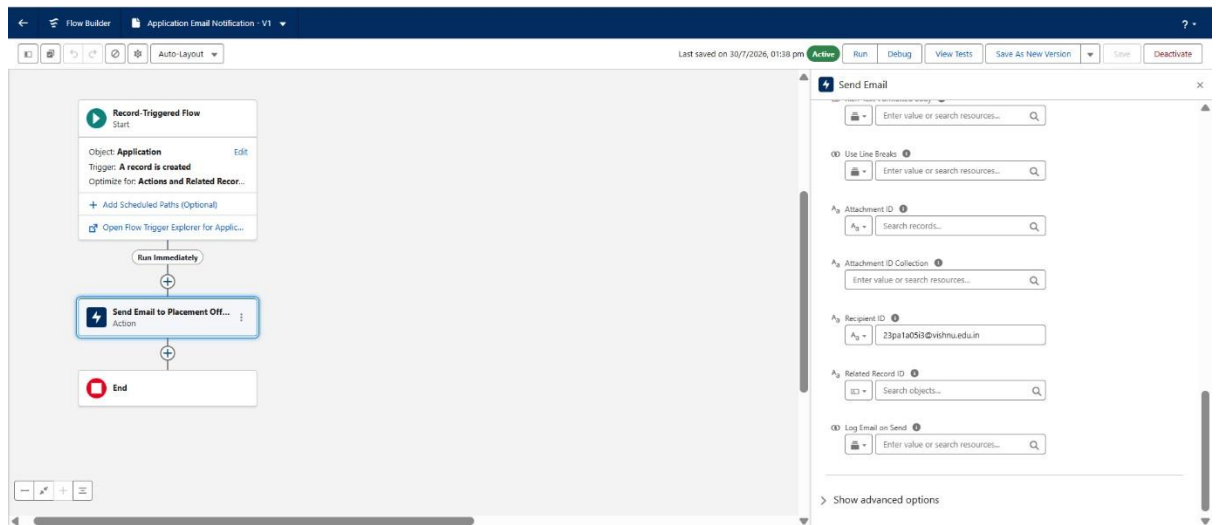
Details	
Application Name	REGD00000000003
Student	Jaya
Job	Salesforce Developer
Application Date	30/07/2026
Status	Applied
Created By	Jaya Sri Padilam, 30/07/2026, 12:01 pm
Last Modified By	Jaya Sri Padilam, 30/07/2026, 12:01 pm

2. Email Notification

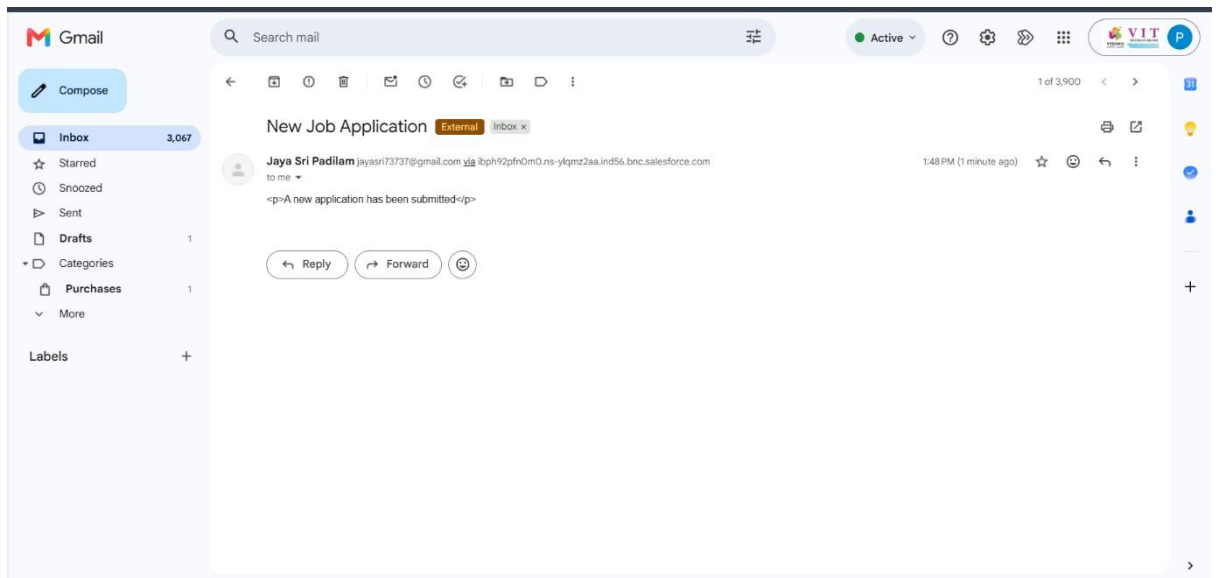
Type: After-Save Record-Triggered Flow

- Object: **Application**
- Trigger: **When a record is created**

- Action: Send Email



The Placement Officer is automatically notified whenever a new application is submitted.



Part 5 – Hands-on Task: Validation Rules

Objective

Create Validation Rules in the Application object to ensure only valid applications are submitted. Validation Rules improve data quality by preventing invalid records from being saved

Validation Rule 1 – CGPA Check

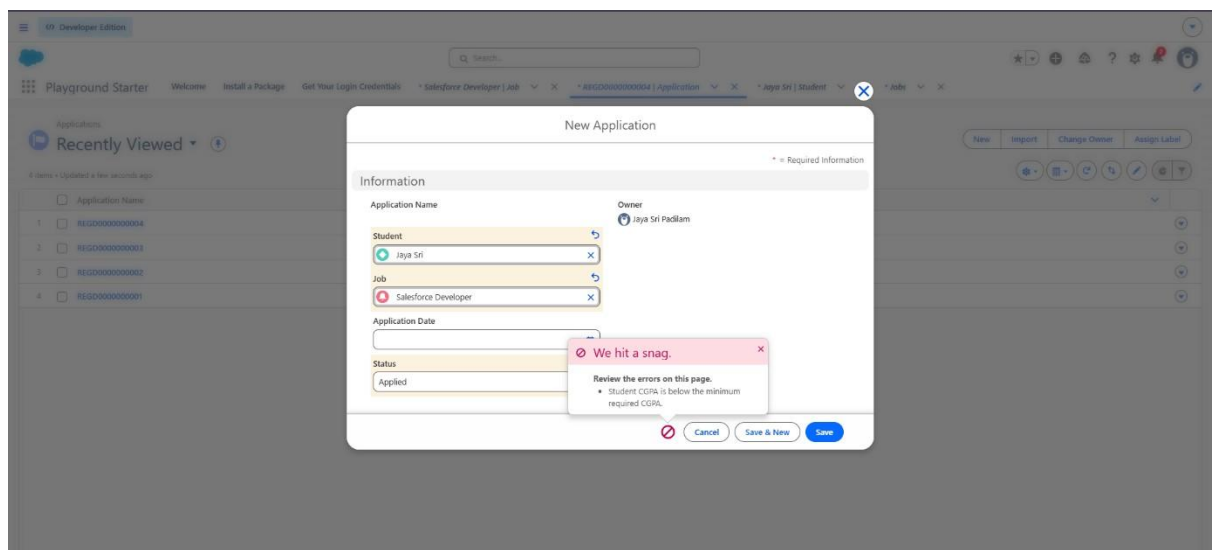
Rule Name: CGPA_Check

Purpose:

Prevent students from applying if their CGPA is less than the minimum CGPA required for the selected job.

Formula:

Student__r.CGPA__c < Job__r.Minimum.CGPA__c



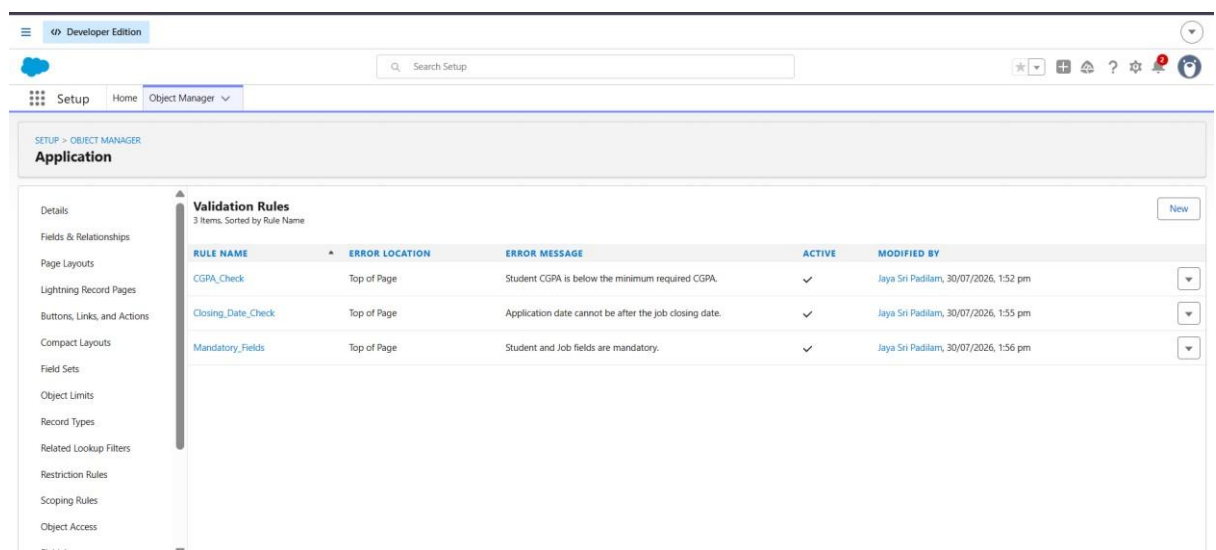
Validation Rule 2 – Closing Date Check

Rule Name: Closing_Date_Check

Purpose:

Prevent students from applying after the job's closing date.

Formula: Application_Date__c > Job__r.Closing_Date__c



Validation Rule 3 – Mandatory Fields

Rule Name: Mandatory_Fields

Purpose:

Ensure that both the Student and Job fields are selected before saving an application.

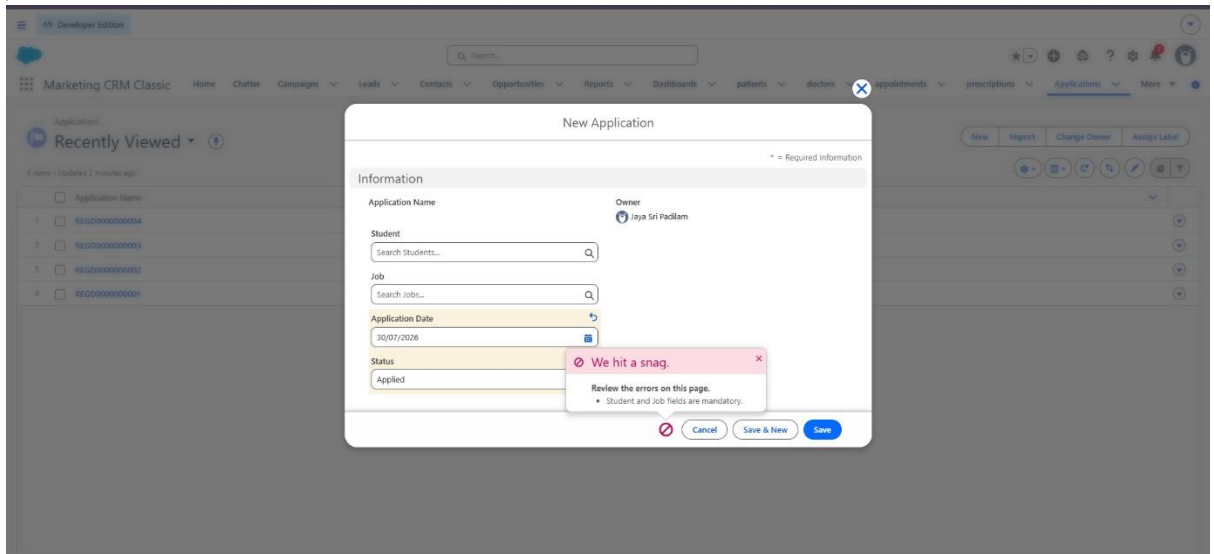
Formula:

OR(

ISBLANK(Student__c),

ISBLANK(Job__c)

)



Part 7 – Mini Project Enhancement

Objective

Enhance the Placement Management System by implementing automation using RecordTriggered Flows and Validation Rules. The system should automate the application process and improve data accuracy.

1. Auto-fill Application Date

Implementation:

- Created a Before-Save Record-Triggered Flow on the Application object.
- Configured the flow to automatically set the Application Date to the current date when a new application is created.

Result:

The Application Date is filled automatically, eliminating manual data entry.

2. Email Notification

Implementation:

- Created an After-Save Record-Triggered Flow.
- Configured the flow to send an email to the Placement Officer whenever a new application is submitted.

Result:

The Placement Officer receives an automatic notification for every new application.

3. Validation Rules

Implemented the following validation rules:

- CGPA Validation: Prevents students from applying if their CGPA is below the job's minimum CGPA.
- Closing Date Validation: Prevents applications after the job closing date.
- Mandatory Fields Validation: Ensures Student and Job fields are not left blank.

Result:

Only valid application records are saved in Salesforce.

4. Automatic Offer Letter Creation

Implementation:

- Created a Record-Triggered Flow on the Application object.
- Configured the flow to run when the Status is updated to Selected.
- Added a Create Records element to create an Offer Letter record automatically.

Result:

The flow was successfully created and activated. However, during testing, the Offer Letter record was not generated, indicating that additional debugging is required.

Part 8 – Debugging Challenge

Scenario

During testing of the Placement Management System, some automation features did not work as expected. The task was to identify the issue, analyze the possible causes, and apply debugging techniques to resolve the problem

Challenge 1 – Offer Letter Flow Not Creating Records

Problem:

The Offer Letter record was not created automatically when the Application Status was updated to Selected.

Possible Causes:

- Incorrect flow entry conditions.
- Required fields in the Offer Letter object were not populated.
- Incorrect field mapping in the Create Records element.
- The flow was not triggered because the record did not meet the entry conditions.

